

Project Design Phase

Solution Architecture

Date:	03 October 2026
Team ID:	SWTID-2026-3257
Project Name:	Implement Client Script & UI Policy (Incident Table)
Maximum Marks:	4 Marks

1. Solution Architecture Overview & Goals

Solution architecture bridges the gap between business problems and technology solutions. For the 'Implement Client Script & UI Policy on Incident Table' project, the architecture defines how dynamic client-side controls interact with the ITSM platform application engine and cloud database to streamline ticket management and maintain strict data governance.

- Find the best technical solution to eliminate data entry errors and form clutter on the Incident table.
- Describe the structure, characteristics, and dynamic behaviors of Client Scripts and UI Policies for project stakeholders.
- Define features, development phases, and solution requirements across ticket lifecycle states (New, In Progress, Resolved, Closed).
- Provide clear specifications according to which the dynamic UI policies and validation scripts are delivered and managed.

2. Architectural Components & Interaction Flow

The solution architecture consists of three primary layers: Presentation Layer (Browser Client), Platform Business Logic Layer (Client Script & UI Policy Engine), and Data Persistence Layer (Cloud DB).

Layer / Component	Role & Description	Technical Stack
Presentation Layer (User Interface)	Service Operations Workspace / Incident Form UI where Service Desk Analysts enter ticket data.	HTML5, CSS3, GlideForm (g_form) API
Client-Side Policy Engine (UI Policies)	Evaluates field conditions declarative-style (e.g., State = Resolved) to toggle field visibility, mandatory status, and read-only attributes.	ServiceNow UI Policy Engine, JavaScript DOM manipulation
Client-Side Execution Engine (Client Scripts)	Listens for field events (onChange, onLoad, onSubmit) to execute real-time calculations (Priority matrix) and user guidance prompts.	JavaScript, GlideForm (g_form), GlideUser (g_user), GlideAjax

Server Application Layer	Processes validated payload submissions, enforces server-side business rules, and triggers automated workflows.	ServiceNow Application Engine, Glide System
Database & Audit Layer	Stores incident records, state logs, assignment histories, and audit trails.	Relational Database Instance (MariaDB / MySQL Enterprise Cloud)

3. End-to-End Incident Lifecycle Architecture Flow

Phase 1: Form Initialization (onLoad)

When an analyst opens an incident record, the onLoad Client Script checks user roles and sets initial field layout, keeping resolution fields hidden.

Phase 2: Triage & Dynamic Calculation (onChange)

As Impact or Urgency is updated, an onChange Client Script auto-recalculates Priority instantaneously without page refresh. If Priority = P1, a UI Policy dynamically requires Work Notes.

Phase 3: Ticket Resolution (State = Resolved)

When State is switched to 'Resolved', a UI Policy triggers to reveal 'Resolution Code' and 'Resolution Notes', marking both as mandatory.

Phase 4: Form Submission (onSubmit)

The onSubmit Client Script verifies all mandatory fields are populated. If valid, data is sent via HTTPS REST API to the platform engine.

Phase 5: Ticket Closure & Read-Only Lock (State = Closed)

Once State reaches 'Closed', a UI Policy sets all core fields to Read-Only, locking the record against unauthorized edits.